"Morris" 5-DOF Robotic Arm

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Contents

1 Introduction

Morris

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2 Bill of Materials (BOM)

Table 1: Purchased Parts

Description	Qty
MG996R Servo	5
ESP32 Devkit Type-C	1
PCA9685 16-Channel 12-Bit PWM Driver	1
DC5525 to Screw Terminal Connector DC Power Jack (Female)	1
10A Adjustable DC-DC Step Down Converter	1

3 Analysis

3.1 Inverse Kinematics

When calculating the inverse kinematics (IK) of the Morris robot we will consider the robot's spatial frame {s} to be placed at the center of the base of the robot and aligned such that the −y direction faces forward, the −x direction faces right, and the +z direction faces upwards.

The n links of the robot are taken as L_n with joints J_n. As link 2 has an L-shaped structure it is more specifically denoted by the horizontal and vertical offsets between the joints J_2 and J_3, notated as L_{2x} and L_2y

The following will cover the IK required to position the robot's end-effector at a point $\vec{p} = (p_x, p_y, p_z)$ in 3D space and an orientation ϕ in the 2D space of the robot's wrist. The robot is kinematically decoupled into three independent systems. The base pivot J_1, the main arm J_2 and J_3, and the wrist J_4.

J_1 is considered a 1R mechanism that acts in the x−y plane that only needs to point towards the target point $\vec{p}$, as such the IK angle for J_1 is found via

$$\theta_1 = \text{atan2}(p_y, p_x)$$

J_2, J_3, and J_4 are all considered coplanar. In order to apply IK to these joints, the point $\vec{p}$ must be converted into a point $\vec{p'} = (p'_x, p'_y)$ this plane. The x coordinate of this point can be found as magnitude of the x−y plane projection of $\vec{p}$; whilst the y coordinate is simply the z coordinate of $\vec{p}$, therefore

$$p'_x = \sqrt{p_x^2 + p_y^2}$$
$$p'_y = p_z$$

J_2 and J_3 form a 2R mechanism that is decoupled from J_4. The IK equations of this 2R mechanism must therefore be calculated for a point $\vec{P} = (P_x, P_y)$ which is offset from $\vec{p'}$ by the length L_4 at the angle ϕ, the coordinates of this point are found as

$$P_x = p'_x - L_4 \cos(\phi)$$
$$P_y = p'_y - L_4 \sin(\phi) - h$$

where h is the vertical offset of J_1 from the origin of {s}.

As link 2 posssesses an L-shaped structure, an "offset angle" θ_0 is established, and the length of the effective link L_2 is calculated from its components,

$$\theta_0 = \text{atan2}(L_{2y}, L_{2x})$$
$$L_2 = \sqrt{L_{2x}^2 + L_{2y}^2}$$